

Cambridge (CIE) IGCSE Physics
Paper 3: Theory (Set A)
Core

Saturday 26 April 2025

Morning (Time: 1 hour 15 minutes)

Total marks

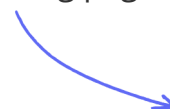
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Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8N (acceleration of free fall = 9.8m/s^2).
- You should show all your working and use appropriate units.

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or visit the mock exams landing page for this course



1 (a) A student reviews some data about athletes and footballers.

An athlete runs 12 km in 1.5 hours.

Calculate the athlete's average speed in km/h.

average speed = km/h

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(3 marks)

(b) Fig. 2.1 shows the speed-time graph for a footballer for the first 15.0 seconds of a game.

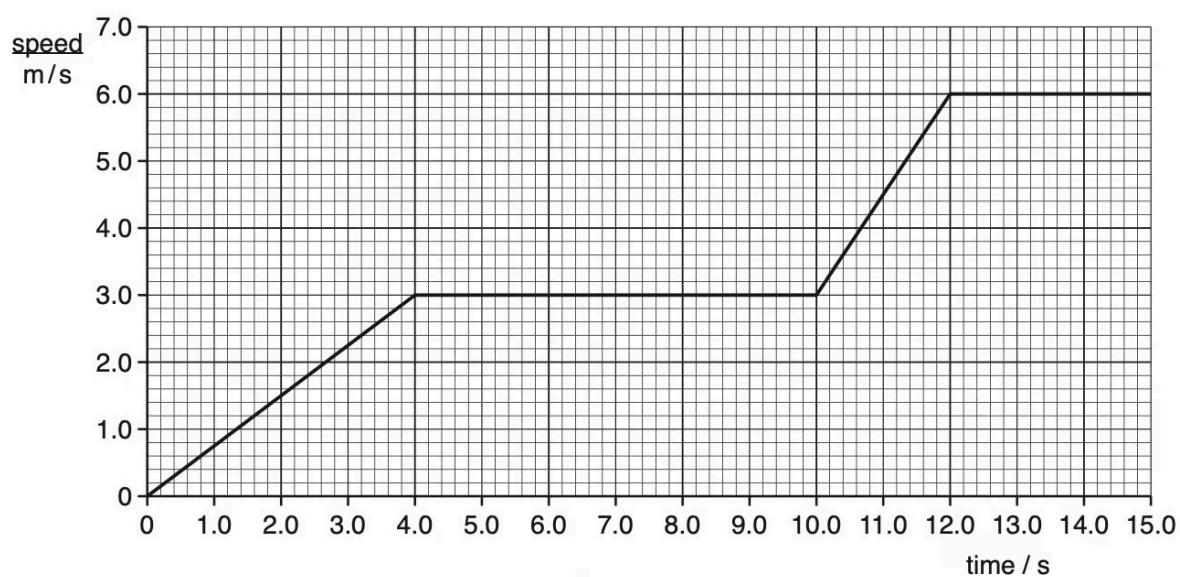


Fig. 2.1

(i) Use the graph in Fig. 2.1 to calculate the distance travelled by the footballer during the first 4.0 seconds.

distance = m [3]

(ii) Use the graph in Fig. 2.1 to determine when the footballer is moving with greatest acceleration.

Between s and s.

Give a reason for your answer.

[2]

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(5 marks)

(c) Another footballer has a mass of 72kg.

Calculate the weight of this footballer.

weight = N

.....

(3 marks)

- 2 (a)** A student has an irregularly shaped piece of metal, a beaker of water and a measuring cylinder, as shown in Fig. 2.1.

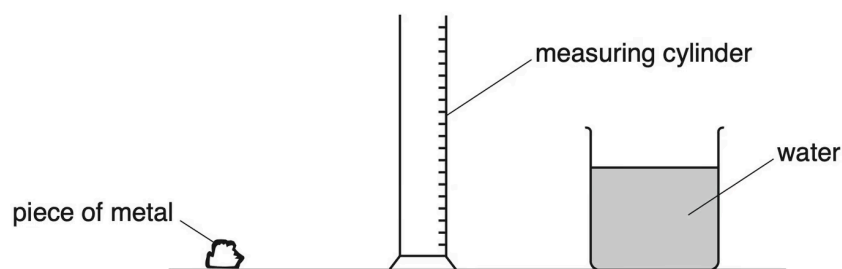


Fig. 2.1

Describe how the student can accurately determine the volume of the piece of metal using the equipment provided.

(4 marks)

- (b)** The student measures the mass of the piece of metal. Its mass is 146 g.

(i) State the name of the instrument used to measure the mass.

[1]

(ii) The volume of the piece of metal is 20 cm³.

Calculate the density of the metal. State the unit.

density = [4]

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(5 marks)

3 (a) Fig. 2.1 shows a man pushing down on a lever to lift one end of a heavy log.



Fig. 2.1

State the term used to describe the turning force exerted by the man.

(1 mark)

(b) (i) Fig. 2.2 shows the forces acting as the man starts to lift the heavy log.

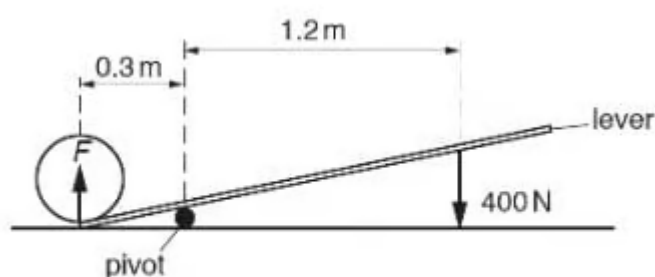


Fig. 2.2

Calculate the force F , exerted by the lever on the heavy log.

force = N [3]

(ii) Describe how the man can use a smaller force to lift the heavy log.

[1]

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(4 marks)

- 4 (a)** A tidal barrage (dam) produces electricity using tides. Fig. 5.1 shows a diagram of a tidal barrage (simplified).

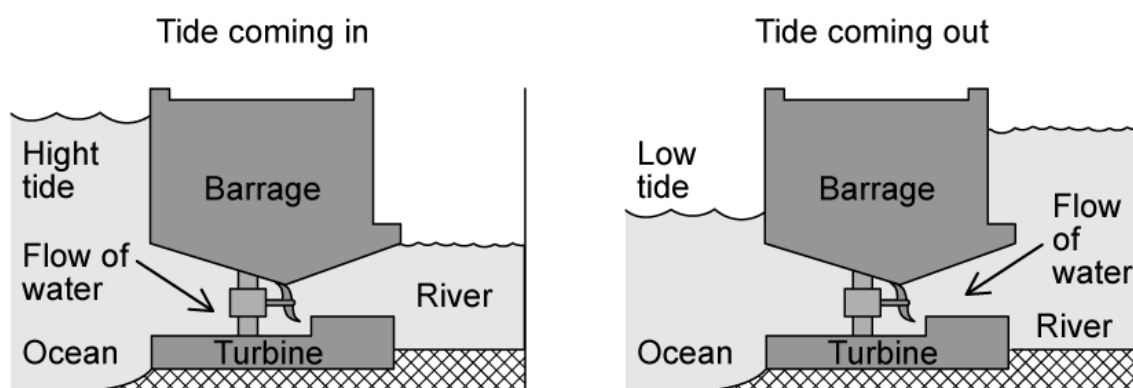


Fig. 5.1

The water behind the barrage (dam) has most of its energy in one of its energy stores. State the name of this energy store.

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(1 mark)

- (b)** Explain how the tidal barrage (dam) produces electricity.

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.....
(3 marks)

- 5 (a)** Fig. 4.1 shows a smoke cell. The cell contains smoke particles and air molecules. It is lit from the side. A student views the motion of smoke particles in the cell by using a microscope.

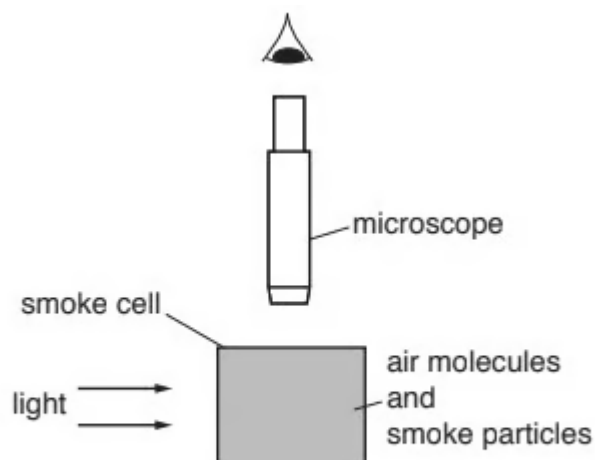


Fig. 4.1

Describe and explain what the student sees when viewing the smoke particles through the microscope.

(4 marks)

- (b)** Drops of water on a warm surface disappear after a short time. State the term used to describe this process. Explain the process, using your ideas about molecules.

name of process:

explanation:

(3 marks)

- 6 (a)** A student investigates the thermal energy lost from two metal cans, X and Y. The cans are identical apart from their outside colour.

The student pours the same volume of hot water into each can and seals each can. The student records the temperature of the water in each can at regular time intervals for a period of 35 minutes.

The equipment is shown in Fig. 6.1.

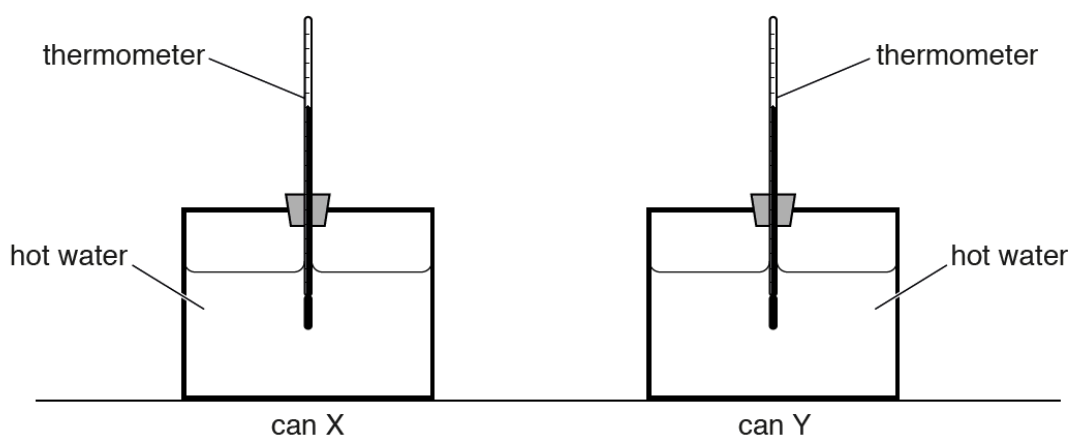


Fig. 6.1

Fig. 6.2 is a graph of the results from the investigation.

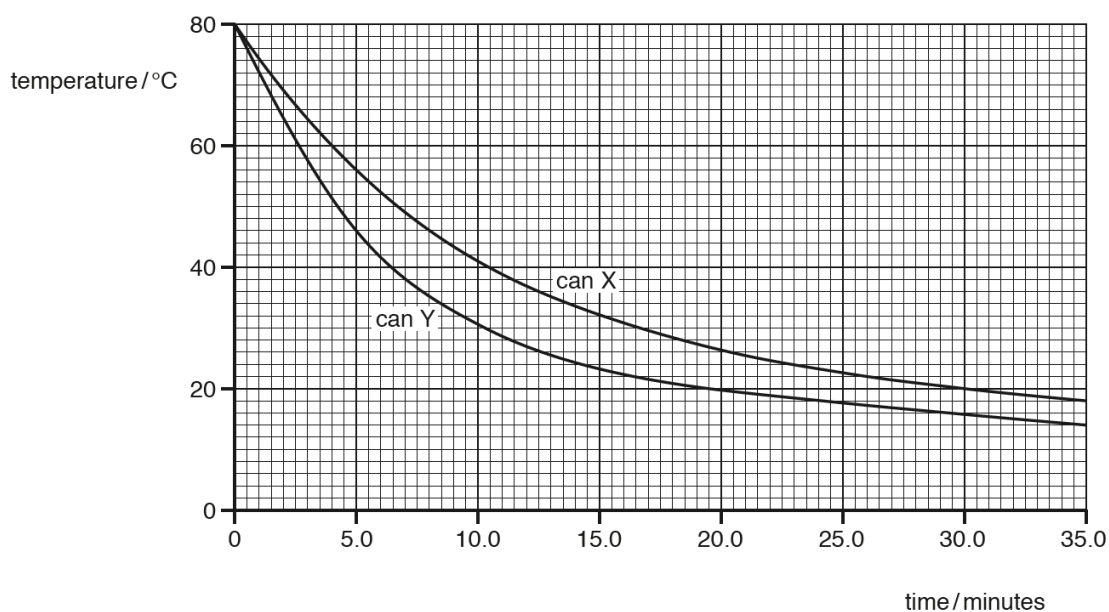


Fig. 6.2

For can X, use Fig. 6.2 to determine the drop in temperature of the water

(i) in the **first** five minutes

[1]

(ii) in the **last** five minutes

[1]

(2 marks)

- (b) Explain why the water cools at a greater rate during the first five minutes of the experiment, compared with the last five minutes.

(2 marks)

- (c) The outside of one can is dull black and the outside of the other is shiny white.

State the colour of can Y.

Explain your answer.

(2 marks)

7 (a) Fig. 8.1 represents the pressure at one instant along part of a sound wave.

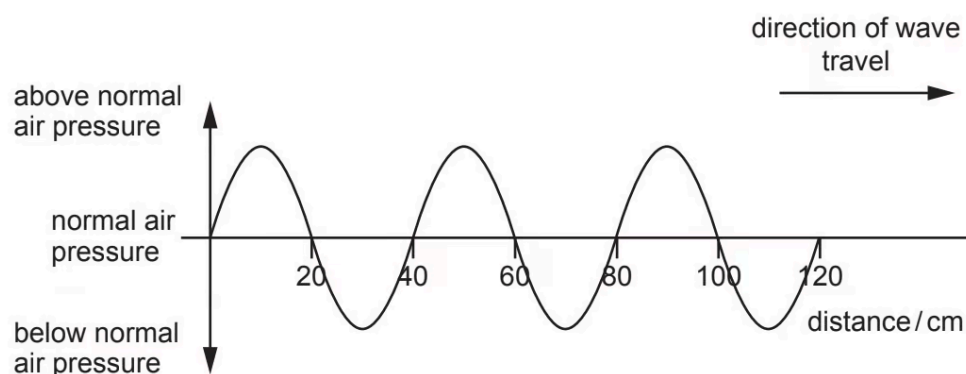


Fig. 8.1

(i) Determine the wavelength of the sound wave.

wavelength of the sound wave = cm [1]

(ii) On Fig. 8.1, draw a wave representing a louder sound of the same wavelength.

[1]

(2 marks)

(b) State the range of audible frequencies for a healthy human ear. Include the unit.

(2 marks)

- 8 (a)** Fig. 6.1 shows a mirror periscope. The periscope is used to view a golfer over the heads of other people. The periscope has two plane mirrors each at an angle of 45° to the vertical.

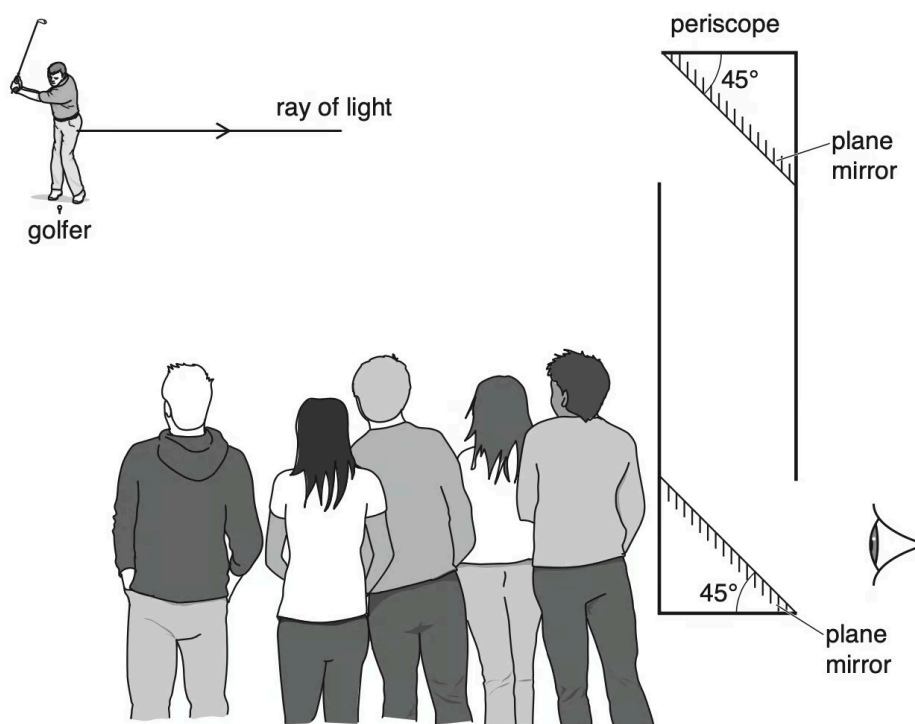


Fig. 6.1 (not to scale)

(i) On Fig. 6.1:

1. Continue the ray of light from the golfer towards the upper mirror of the periscope.

- Draw and label the normal at the point where the ray strikes the mirror.

[1]

(ii) On Fig. 6.1, continue the ray of light after reflection at the upper mirror until it leaves the periscope.

[1]

(iii) State the law of reflection used to deduce the position of the ray of light after striking the mirrors.

[1]

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(3 marks)

(b) Fig. 6.2 shows three rays of red light each entering a semi-circular glass block.

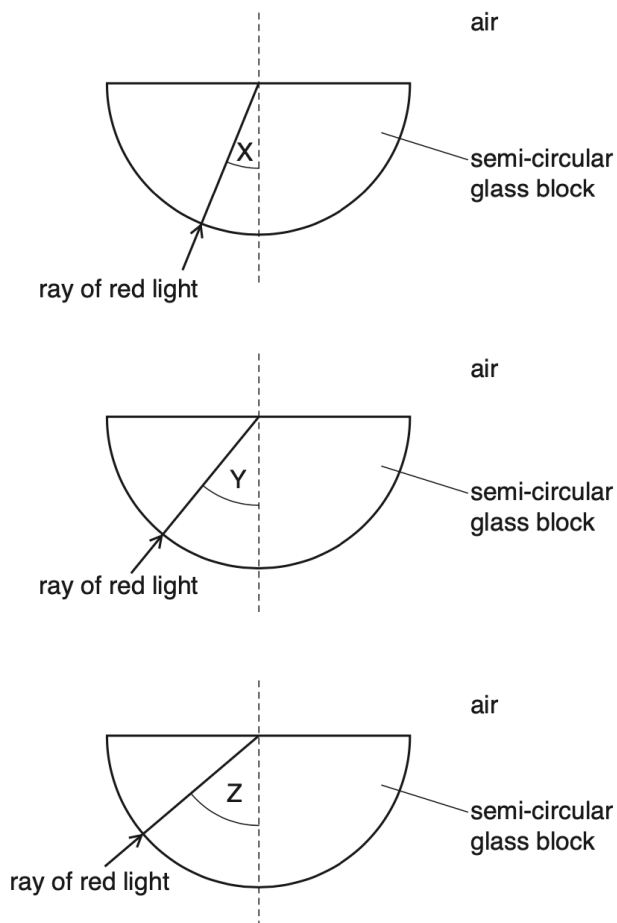


Fig. 6.2

Table 6.1

angle of incidence	description
X	less than the critical angle
Y	equal to the critical angle
Z	greater than the critical angle

Using the information in Table 6.1, draw on Fig. 6.2 to complete the path of each ray of red light.

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(3 marks)

9 (a) Fig. 7.1 shows some parts of the electromagnetic spectrum.

radio waves		infra-red waves	visible light	ultraviolet waves	X-rays	γ -rays
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Fig. 7.1

(i) In Fig. 7.1, one part of the electromagnetic spectrum is not labelled.

State the name of this part.

[1]

(ii) The speed of visible light waves in a vacuum is 3.0×10^8 m / s.

Suggest a value for the speed of infra-red waves in a vacuum.

speed = m / s

[1]

(iii) Some parts of the electromagnetic spectrum have a wavelength shorter than that of visible light. State one example.

[1]

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.....

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(3 marks)

(b) (i) X-rays and γ -rays are used in hospitals.

Describe one medical use for X-rays and one use for γ -rays.

[1]

(ii) Explain why γ -rays are dangerous to living things.

[1]

(4 marks)

10 (a) Fig. 10.1 shows the symbol for an electrical component.



Fig. 10.1

State the name of the component shown in Fig. 10.1.

(1 mark)

(b) The resistance of the component shown in Fig. 10.1 varies with temperature. Fig. 10.2 shows a graph of resistance against temperature for the component.

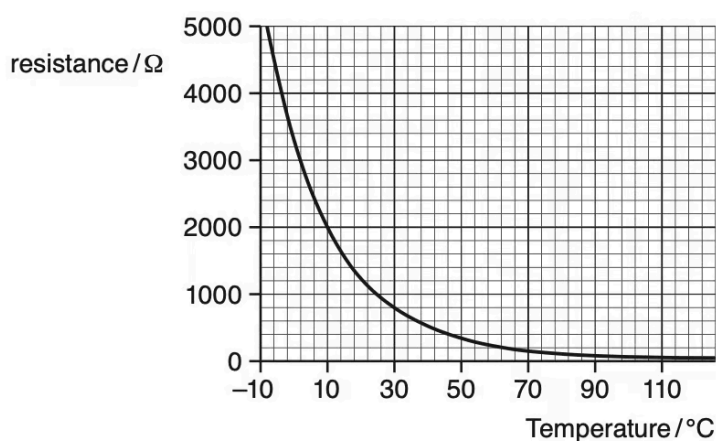


Fig. 10.2

(i) Use the graph to determine the resistance of the component at a temperature of 10 °C.

resistance = Ω [1]

(ii) At another temperature, the resistance of the component is 800 Ω.

Calculate the current in the component when it is connected to a 12.0 V supply.

current = A [3]

(4 marks)

- (c)** The students who built the circuit in part (a) want to change it so that it becomes sensitive to light rather than heat.

State the change the students should make.

(2 marks)

- (d)** Draw the symbol for the replacement component you suggested in part (c).

(1 mark)

11 (a) Fig. 12.1 shows a diagram to represent a helium atom, and an incomplete key.

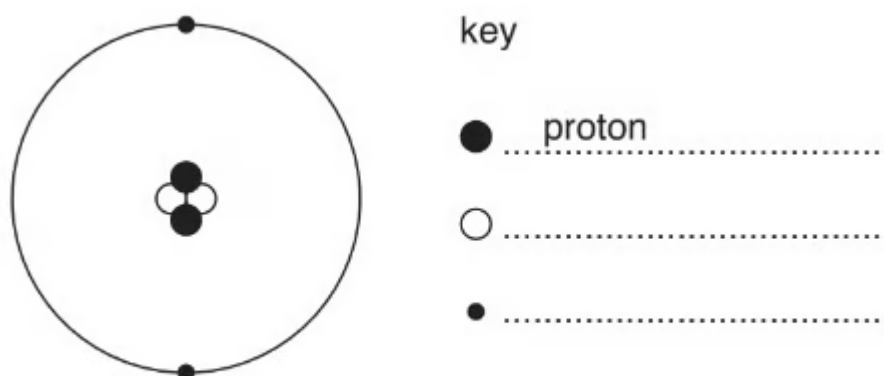


Fig. 12.1

Complete the key in Fig. 12.1. State the name of each particle.

.....

.....

(2 marks)

(b) The table in Fig. 12.2 compares two isotopes of helium.

	${}^3_2\text{He}$	${}^5_2\text{He}$
number of protons		
number of neutrons		

Fig 12.2

For each isotope, write the number of protons and the number of neutrons in the correct places in the table.

(2 marks)

(c) The nucleus of the helium atom in (a) is the same as an α -particle.

(i) Describe the penetrating ability of α -particles.

[1]

(ii) Explain why it is dangerous to swallow a source that emits α -particles.

[2]

(3 marks)

12 (a) Fig. 1 shows a diagram of the solar system.

For each position (i - iii) state the name of the body indicated.

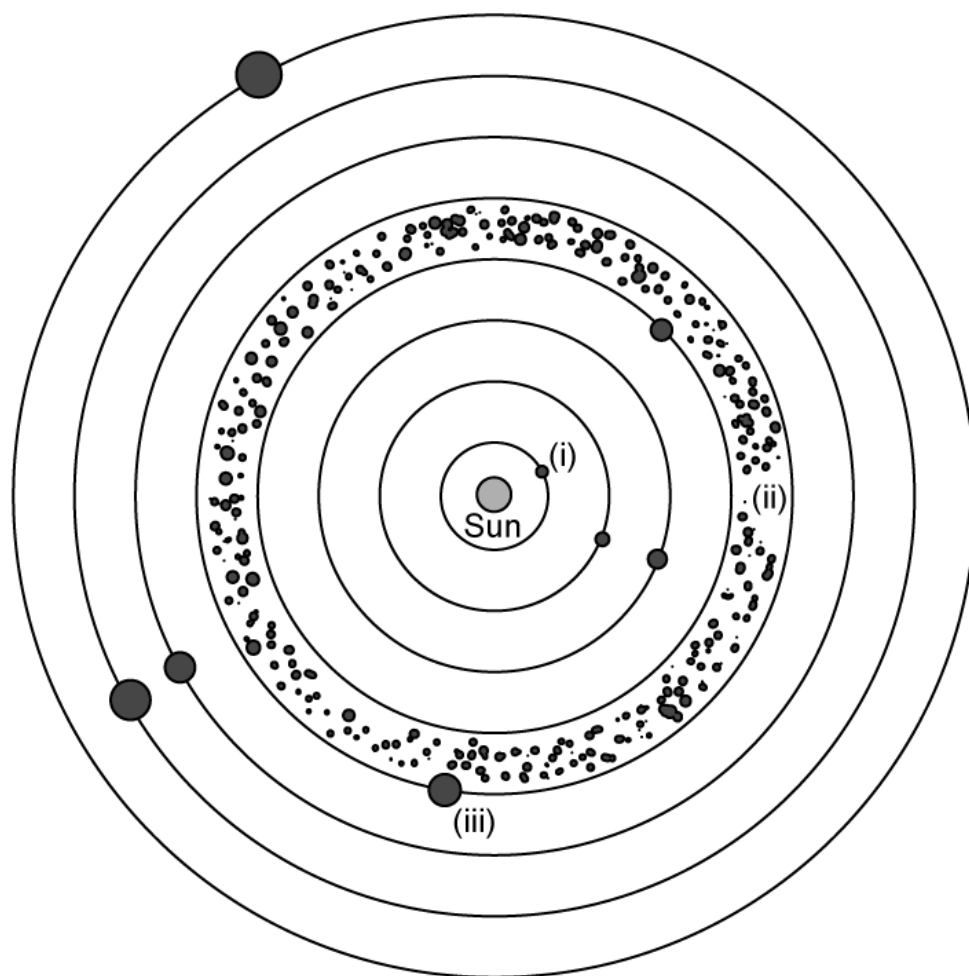


Fig. 1 (not to scale)

(3 marks)

(b) There are more than 200 moons in our solar system.

(i) State the meaning of the term 'moon'.

[1]

(ii) Choosing from the labelled bodies in part (a), state the body or bodies which have moons.

[1]

(iii) Earth has one moon, known as the Moon. Name one other planet in the solar system apart from the answer to part (ii) which has a moon.

[1]

(3 marks)